Projectile Motion

Projectile motion is the motion of an object thrown or projected into the air, subject to ***only*** the acceleration of gravity. The object is called a ***projectile***, and its path is called its ***trajectory***. The motion of a falling object, as we discussed before, is a simple one-dimensional type of projectile motion in which there is no horizontal movement.

In this chapter, we consider two-dimensional projectile motions, such as horizontal projectile and projectile launched with an angle. In our projectile motion studies, ***air resistance is always negligible*** unless otherwise specified.

The Independence of Perpendicular Motions

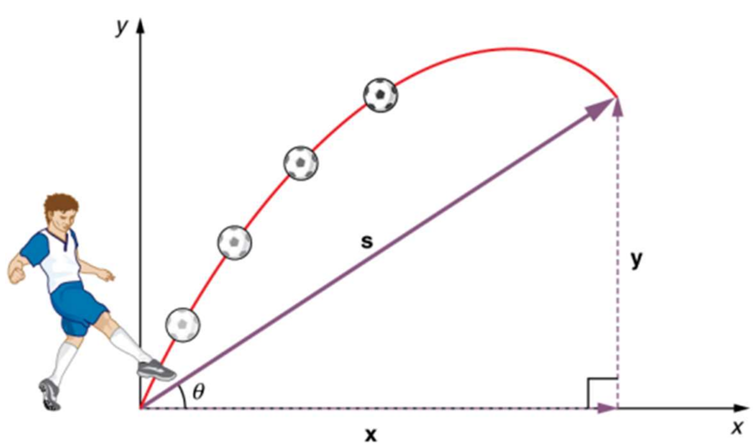
The horizontal and vertical components of two-dimensional motion are independent of each other. Any motion in the horizontal direction does not affect motion in the vertical direction, and vice versa.

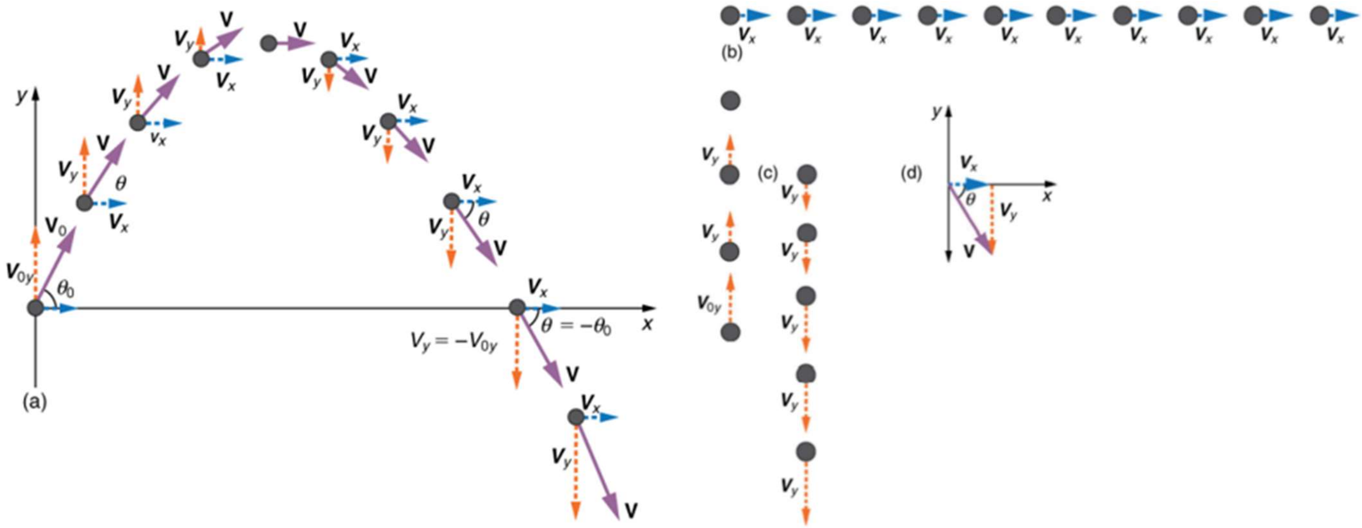
The Connection of Components of a Motion

The horizontal and vertical components of a motion take exactly the ***same amount of time***, since they both are from the same motion. ***Since motions along perpendicular axes are independent, they can be analyzed separately.***

Table 3.1 Equations of Kinematics for Constant Acceleration in Two-Dimensional Motion

x Component	Variable	y Component
x	Displacement	y
a_x	Acceleration	a_y
v_x	Final velocity	v_y
v_{0x}	Initial velocity	v_{0y}
t	Elapsed time	t
$v_x = v_{0x} + a_x t$		$v_y = v_{0y} + a_y t$
$x = \frac{1}{2}(v_{0x} + v_x)t$		$y = \frac{1}{2}(v_{0y} + v_y)t$
$x = v_{0x}t + \frac{1}{2}a_x t^2$		$y = v_{0y}t + \frac{1}{2}a_y t^2$
$v_x^2 = v_{0x}^2 + 2a_x x$		$v_y^2 = v_{0y}^2 + 2a_y y$

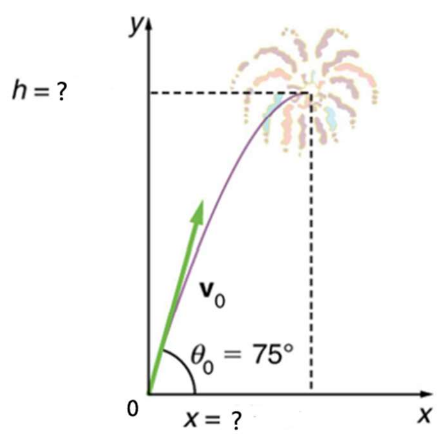




Example 1: 斜抛 Projectile Launched with an Angle

A Fireworks Projectile Explodes High and Away

During a fireworks display, a shell is shot into the air with an initial speed of 70.0 m/s at an angle of 75.0° above the horizontal, as illustrated in Figure 3.38. The fuse is timed to ignite the shell just as it reaches its highest point above the ground. (a) Calculate the height at which the shell explodes. (b) How much time passed between the launch of the shell and the explosion? (c) What is the horizontal displacement of the shell when it explodes?



y Component	*x* Component
$v_y = v_{0y} + a_y t$	
$y = \frac{1}{2}(v_{0y} + v_y)t$	
$y = v_{0y}t + \frac{1}{2}a_y t^2$	
$v_y^2 = v_{0y}^2 + 2a_y y$	

Defining a Coordinate System

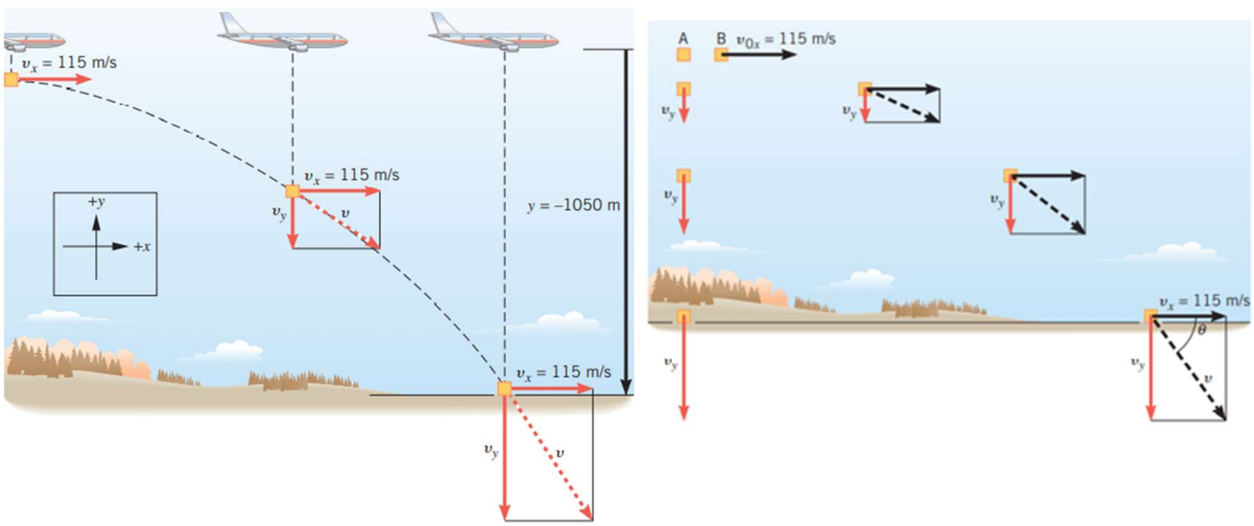
It is important to set up a coordinate system when analyzing projectile motion. One part of defining the coordinate system is to define an origin for the x and y positions. Often, it is convenient to choose the initial position of the object as the origin such that $x_0 = 0$ and $y_0 = 0$. It is also important to define the positive and negative directions in the x and y directions.

Example 2: 平抛

Figure 3.7 shows an airplane moving horizontally with a constant velocity of +115 m/s at an altitude of 1050 m. The directions to the right and upward have been chosen as the positive directions. The plane releases a "care package" that falls to the ground along a curved trajectory. Ignoring air resistance, determine the time required for the package to hit the ground.

2. What is the horizontal displacement of the package once it hits the ground?

3. What is the speed of the package right before it hits the ground?



y Component	x Component

$$v_y = v_{0y} + a_y t$$

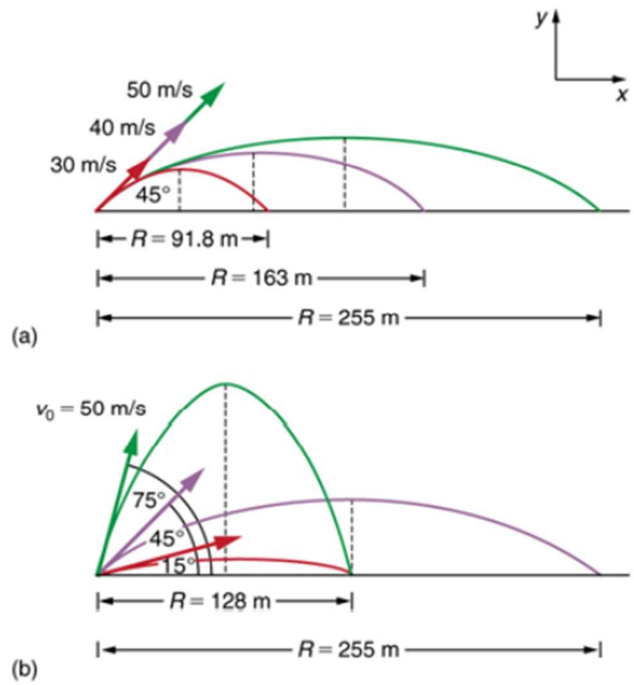
$$y = \tfrac{1}{2}(v_{0y} + v_y)t$$

$$y = v_{0y}t + \tfrac{1}{2}a_y t^2$$

$$v_y^2 = v_{0y}^2 + 2a_y y$$

Range：射程

One of the most important things illustrated by projectile motion is that vertical and horizontal motions are independent of each other. Galileo was the first person to fully comprehend this characteristic. He used it to predict the range of a projectile. On level ground, we define **range** to be the horizontal distance R traveled by a projectile. Galileo and many others were interested in the range of projectiles primarily for military purposes—such as aiming cannons. However, investigating the range of projectiles can shed light on other interesting phenomena, such as the orbits of satellites around the Earth. Let us consider projectile range further.



$$R = \frac{v_0^2 \sin 2\theta_0}{g},$$

How does V0 affect range? How does the angle θ_0 of initial velocity affect range?

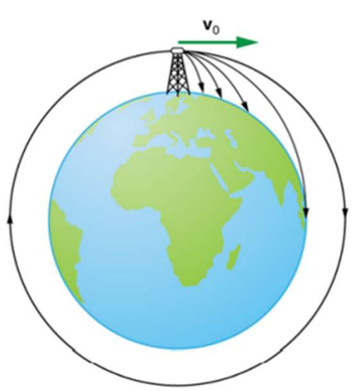


Figure 3.41 Projectile to satellite. In each case shown here, a projectile is launched from a very high tower to avoid air resistance. With increasing initial speed, the range increases and becomes longer than it would be on level ground because the Earth curves away underneath its path. With a large enough initial speed, orbit is achieved.

Example 3：射程

25. A projectile is launched at ground level with an initial speed of 50.0 m/s at an angle of 30.0° above the horizontal. It strikes a target above the ground 3.00 seconds later. What are the x and y distances from where the projectile was launched to where it lands?

平抛

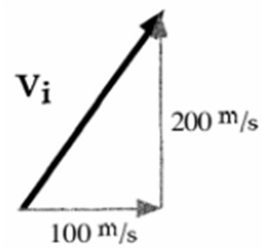
27. A ball is thrown horizontally from the top of a 60.0-m building and lands 100.0 m from the base of the building. Ignore air resistance. (a) How long is the ball in the air? (b) What must have been the initial horizontal component of the velocity? (c) What is the vertical component of the velocity just before the ball hits the ground? (d) What is the velocity (including both the horizontal and vertical components) of the ball just before it hits the ground?

19. mmh A golfer imparts a speed of 30.3 m/s to a ball, and it travels the maximum possible distance before landing on the green. The tee and the green are at the same elevation. **(a)** How much time does the ball spend in the air? **(b)** What is the longest hole in one that the golfer can make, if the ball does not roll when it hits the green?

Estimating the Projectile Motion with a Launch Angle

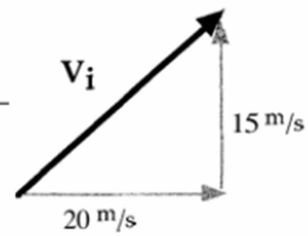
Estimate numerical answers using $g = 10\ m/s^2$.

1. A rocket is fired over level ground. The components of the initial velocity are given.
   a) Which component determines how long it will be in the air, v_x or v_{yi}? _______
   b) How long is the rocket in the air? _______
   c) How far away does it land? _______

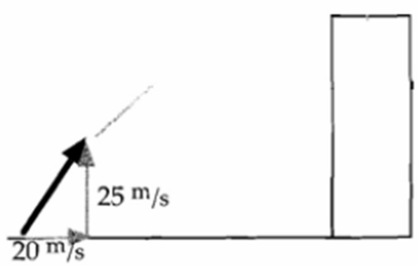


2. Two projectiles are fired on level ground. One ball is fired straight up, and one ball is fired at an angle. The balls reach the same height. What else is the same? Circle all correct responses.
   A. v_x
   B. v_{yi}
   C. v_f total
   D. time in air, t

3. A football is kicked over a level field. The initial velocity components are given.
   a) How fast is it going when it reaches the highest point in its trajectory? _______
   b) What is its speed just before it hits the ground? _______
   c) How far away does it land? _______



4. Now the same football is kicked up onto a roof. It is in the air for 3.0 seconds. The initial velocity components are given.
   a) What is v_{xf}? _______
   b) What is v_{yf}? _______
   c) What is its final speed? _______



d) Sketch the v-t graph of y-component velocity from 0-3s.

e) Sketch the a-t graph of the football from 0-3s.

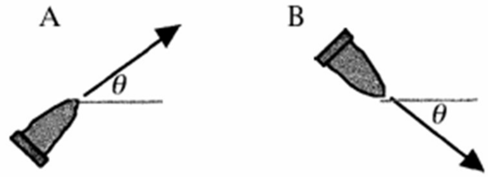
5. A gun always fires a bullet at the same initial speed (known as the muzzle velocity) in any direction. Gun A is aimed at $\theta°$ above horizontal. Gun B is aimed at $\theta°$ below horizontal from the same height.

   a) Which gun's bullet hits the ground sooner? _______

   A. B. C. Tie

   b) Which gun's bullet is traveling faster when it hits? _______

   A. B. C. Tie



6. A rocket is launched over level ground. It is in the air for 8.0 s and lands 400 m away.

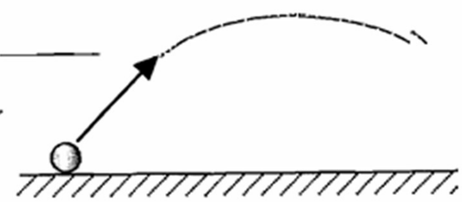
   a) What is v_x? _______

   b) At what time does it reach its maximum height? _______

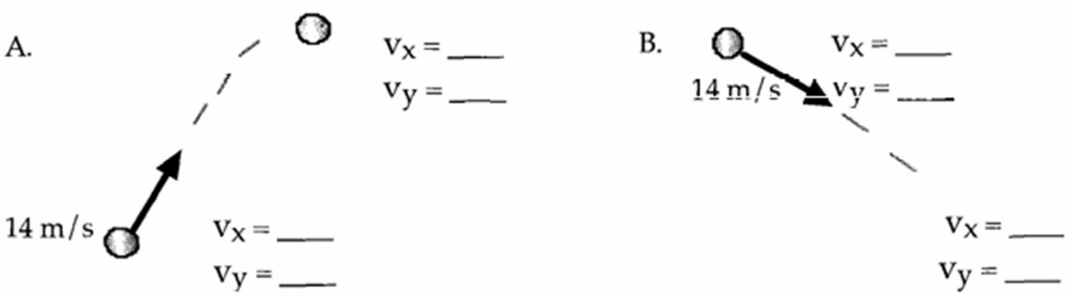
   c) What is v_{yf}? (Hint: Cut problem in half.) _______

   d) How high does it get? _______

   e) What is its final speed? _______



7. Projectile A is launched with a velocity of $v = 14$ m/s, 45° above horizontal. Projectile B is launched with a velocity of $v = 14$ m/s, 45° below horizontal. Sketch and list the velocity components at times $t = 0$ s and $t = 1$ s.

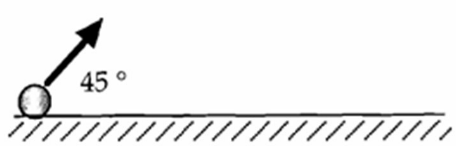


8. A ball is thrown at a 45° angle over level ground. Which statement below is true?

   A. $x_f = y_{max}$ (maximum height)

   B. $x_f > y_{max}$

   C. $x_f < y_{max}$



9. Sketch the a-t graph for the rocket from 0-8s in #6 above. Also sketch the v-t graphs for both Vx and Vy from 0-8s.